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Project report

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Automatic generation of Slovenian traffic news for RTV Slovenija

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Abstract

Traffic reporting at RTV Slovenija is currently a manual process that relies on student workers compiling structured reports from the *promet.si* portal for bi-hourly broadcasts. This workflow is time-consuming and susceptible to human error. In this project, we investigate the potential of automating traffic news generation using large language models (LLMs). We explore several multilingual LLMs and evaluate their suitability for generating structured Slovenian traffic reports based on real-world data. Our methodology involves data preprocessing, prompt engineering techniques such as Role-Based and Few-Shot Prompting, and aligning data from structured traffic logs with their corresponding `.rtf` reports. A key challenge arises from inconsistencies between these two data sources, likely due to the use of additional, undocumented inputs in the original report generation. We describe our pipeline for linking and preprocessing the data, propose a framework for automatic evaluation, and outline steps for future work focused on improving data alignment and testing advanced prompting strategies with the GaMS language model.

Keywords

traffic reports, natural language processing, large language models, prompt engineering

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Introduction

Traffic reporting is a crucial component of public broadcasting, providing real-time updates on road conditions, accidents, and congestion. At RTV Slovenija, traffic news is currently compiled manually by students who review traffic reports from the *promet.si* portal and transcribe them into structured news segments, which are then broadcast every 30 minutes. This manual approach is time-consuming, prone to human error, and inefficient given the volume of traffic data that needs to be processed.

With the advancement of large language models (LLMs), there is an opportunity to automate the generation of structured and readable traffic news. This project aims to leverage an existing LLM, apply prompt engineering techniques and fine-tune the model to automatically generate traffic reports that adhere to the broadcasting standards of RTV Slovenija. In the next chapter, we present various existing solutions that we consider using in this project.

Related Work and Existing Solutions

In their master's thesis, Bjørgo and Lundegaard explored the automation of traffic incident reporting. They developed a system that collects real-time traffic data, processes it to identify incidents, and automatically generates reports for dissemination to the public and relevant authorities. Their work provides a detailed description of approaches that could be highly beneficial to our project [1].

Methods

LLM Selection

For this project, we focus on two large language models (LLMs): *GaMS-9B-Instruct* and *Gemini-2.0-flash*. *GaMS-9B-Instruct* is based on Google's Gemma 2 model family and was pretrained on Slovene and English corpora. *Gemini-2.0-flash*, developed by Google DeepMind, is part of a broader suite of advanced AI models pretrained on multilingual datasets. For simplicity, we will refer to this model as *Gemini*.

Data Analysis

We analyzed the dataset contained in the Excel file TRAFFIC_REPORTS_2022_2023_2024.XLSX, obtained from the *promet.si* portal. The file comprises a total of 170,823 traffic reports from the years 2022, 2023, and 2024, organized across 27 attributes, each accompanied by a corresponding description.

Due to a significant amount of missing or irrelevant data, we reduced the number of attributes to the following:

- `Datum`
- `A1`, `B1`
- `ContentPomembnoSLO`
- `ContentNesreceSLO`
- `ContentZastojiSLO`
- `ContentVremeSLO`
- `ContentOvireSLO`
- `ContentDeloNaCestiSLO`
- `ContentOpozorilaSLO`
- `ContentMednarodneInformacijeSLO`
- `ContentSplosnoSLO`

Based on this structured subset of data, the language model (LLM) is expected to generate the final traffic report.

Our second dataset consists of reference reports. It includes 28,037 `.rtf` files, each containing a desired output report produced by RTV Slovenija. These reports were collected on a monthly basis over the years 2022, 2023, and 2024.

Data Preprocessing

We introduced a new attribute in the Excel file, RTF_file_name, which enables each traffic record to be linked to its corresponding `.rtf` file based on the report date.

The dataset obtained from *promet.si* required additional preprocessing, as many of its fields contained embedded HTML content. Similarly, the reports extracted from the `.rtf` files were converted into a structured `DataFrame` format to allow alignment with the traffic data and to facilitate automated prompt generation.

To ensure the correctness of the file-linking process, we additionally extracted the names of cities and villages from the `.rtf` reports using *classla* tokenization and a predefined list of all city names in Slovenia. We then verified whether these locations appeared in the corresponding traffic data records.

For model training and evaluation, the dataset was partitioned into training (70%), testing (15%), and validation (15%) subsets. Since each entry in the *promet.si* dataset corresponds directly to a specific `.rtf` file, it was sufficient to split only the set of `.rtf` files accordingly.

Prompt Engineering

We use the *Few-Shot Prompting* approach for traffic report generation, leveraging structured traffic data paired with human-written examples to guide the model in producing coherent and contextually appropriate outputs.

A key aspect is the prioritization of events based on urgency. The model follows a defined event hierarchy, from most to least critical:

Wrong-way driver, **Closed motorway**, **Accident with traffic jam on the motorway**, **Traffic jams due to roadworks (short delays often cause rear-end collisions)**, **Main/regional road closure due to an accident**, **Other accidents**, **Broken-down vehicles blocking lanes**, **Animal on the roadway**, **Object or spilled cargo on the motorway**, **Risky roadworks (e.g., closed lanes near/in tunnels)**, and **Traffic jams before Karavanke tunnel or border crossings**.

Few-shot Prompting

For few-shot prompting, we created example inputs by combining structured traffic data with corresponding `.rtf` reports. However, in many cases, the content of the two sources **did not align**.

To investigate this mismatch, we manually compared several entries using report dates and found no errors in our linking process. Instead, we concluded that the `.rtf` reports were likely created using additional, untracked data sources.

We ultimately provided the model with structured inputs and three example prompt–response pairs based on human-written reports from the `.rtf` files.

Fine-tuning

To fine-tune the language model for our specific task, we utilized the *GaMS-9B-Instruct* model in combination with the Low-Rank Adaptation (LoRA) technique. LoRA enables efficient parameter updates by introducing a small set of trainable parameters into the model architecture. This significantly reduces both computational and memory requirements, while still maintaining strong performance on downstream tasks.

We applied supervised fine-tuning (SFT) to adapt the model for traffic report generation. To facilitate this, we constructed a custom dataset composed of prompt-response pairs. Each prompt was generated by converting structured traffic data into a natural language format, while the corresponding response was extracted from human-written reference reports (`.rtf` files). This approach allowed the model to learn how to produce fluent, contextually relevant traffic reports from structured input data.

Two versions of the fine-tuned model were developed. The first was trained on the original dataset, consisting of structured traffic data paired with human-written reports, as described earlier. The second model was trained on a synthetic dataset, in which the responses were generated using Google's *Gemini* language model, based on the same structured inputs. This dual setup enabled a comparative evaluation of models trained on human-written versus AI-generated outputs, helping to assess the viability of synthetic data for supervised fine-tuning in this application domain.

Evaluation

For evaluation, we selected *BERTScore*, a metric that assesses the similarity between two texts based on their semantic content. In our case, it compares the reference report with the report generated by the large language model (LLM). We chose this metric because it utilizes contextual embeddings from pretrained language models, enabling it to capture nuanced semantic similarities beyond simple lexical overlap. As part of the evaluation, we report the BERTScore precision, recall, and F1 score.

Results

After analyzing the dataset, we decided to remove attributes that were either irrelevant or consistently empty. These attributes did not contribute meaningful information for traffic report generation. We retained only the attributes that contained substantial content and were largely non-empty. As illustrated in the figure 1 below, several attributes were clearly irrelevant or sparsely populated.

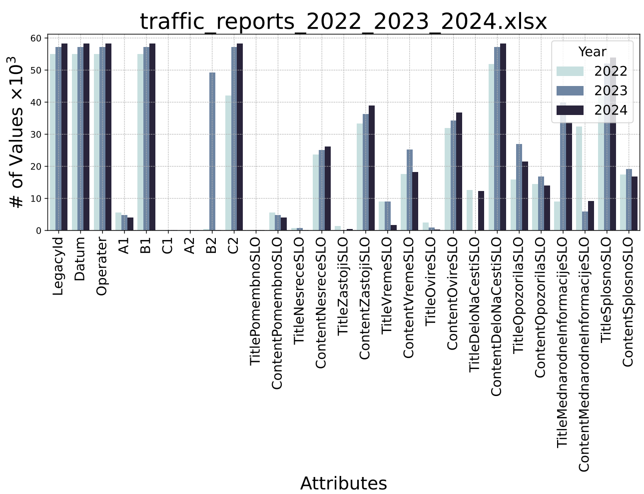

Figure 1. Project's primary dataset visualization.

After fine-tuning several models, we evaluated their performance on the test dataset using *BERTScore*, which provides precision, recall, and F1 score metrics. As shown in Table 1, *Gemini* outperformed *GaMS-9B-Instruct*, which aligns with our expectations given *Gemini*'s larger size and greater number of parameters.

A noteworthy observation is that both fine-tuned models performed worse than the base *GaMS-9B-Instruct* model. We attribute this to the poor quality of the training data, which often contained irrelevant information or lacked critical details that were present in the corresponding human-written reports but missing from the structured input. As a result, the model struggled to learn the appropriate relationships between the input features and the expected output.

Model	Precision	Recall	F1 Score
Gams-9B-Instruct	0.6195	0.6604	0.6387
Gams-9B + SFT (Human)	0.6076	0.6397	0.6227
Gams-9B + SFT (AI)	0.6200	0.6575	0.6375
Gemini	**0.7013**	**0.7397**	**0.7196**

Table 1. Evaluation of LLM models.

Another issue we encountered was that the fine-tuned models tended to mimic the structure of the input data in their outputs, despite being provided with explicit instructions on how the generated reports should be formatted.

Interestingly, the model fine-tuned on AI-generated reports (produced by *Gemini*) outperformed the one trained on human-written reports. This suggests that the AI-generated responses provided a more consistent mapping between the input data and the expected output, enabling the model to learn the generation task more effectively.

Conclusion

Throughout the course of this project, we observed that data analysis and preprocessing are among the most critical components of training large language models (LLMs). Poor data quality can significantly degrade model performance, as evidenced by the inferior results of fine-tuned models compared to the base *GaMS-9B-Instruct* model. To mitigate this, the dataset should be enhanced by incorporating additional relevant information from external sources.

Prompt engineering also plays a crucial role in the training process. Using well-structured input data aligned with compatible human-written reports can lead to substantial improvements in model output quality.

As expected, *Gemini* outperformed *GaMS-9B-Instruct*, likely due to its larger model size and higher number of parameters. This demonstrates that leveraging stronger LLMs such as *Gemini*, can enhance the training process of other models by generating high-quality synthetic data for supervised fine-tuning.

Currently, large language models (LLMs) like *Gemini*, *ChatGPT*, and *DeepSeek* outperform Slovenian models such as *GaMS-9B-Instruct*. Fine-tuning these models could lead to better, more consistent results and has the potential to replace human workers in certain tasks.

References

[1] Anna Jansdatter Bjørgo and Mads Lundegaard. Automating the process of traffic incident reporting. Master's thesis, NTNU, 2024.